

Simulation of FDMA in MATLAB

1. Aim

To simulate a Frequency Division Multiple Access (FDMA) system using MATLAB and analyze its communication performance in terms of channel bandwidth, data rate, and signal-to-noise ratio (SNR).

2. Objectives

- To study hybrid beamforming performance.
 - To analyze the effect of RF chains and beam direction.
 - To study power allocation and SNR.
 - To evaluate the system throughput.
-

3. Software Used

MATLAB

4. Parameters Used

- Channel Bandwidth (kHz)
- Data Rate (kbps)
- Signal-to-Noise Ratio (SNR) (dB)

5. MATLAB Code

```
clc;
clear;
close all;

% Sampling parameters
fs = 1000;
t = 0:1/fs:1;

% FDMA Parameters
TBW = 600;    % Total Bandwidth (kHz)
Users = 3;    % Number of users
CBW = TBW/Users; % Channel bandwidth per user

% Carrier Frequencies
fc = [100 300 500];

% Generate User Signals
m = sin(2*pi*10*t);

s1 = m .* cos(2*pi*fc(1)*t);
s2 = m .* cos(2*pi*fc(2)*t);
s3 = m .* cos(2*pi*fc(3)*t);

% FDMA Signal
fdma = s1 + s2 + s3;

%% Objective 1: Channel Allocation
figure;

subplot(3,1,1)
plot(t,fdma,'b')
grid on
title('FDMA Channel Allocation')
xlabel('Time (s)')
ylabel('Amplitude')

subplot(3,1,2)
bar(1:Users,CBW*ones(1,Users))
grid on
xlabel('Users')
ylabel('Channel Bandwidth (kHz)')
title('Channel Bandwidth per User')

%% Objective 2: Data Rate
DataRate = CBW*2; % kbps
```

```
subplot(3,1,1)
plot(t,fdma,'b')
grid on
title('FDMA Channel Allocation')
xlabel('Time (s)')
ylabel('Amplitude')
```

```
subplot(3,1,2)
bar(1:Users,CBW*ones(1,Users))
grid on
xlabel('Users')
ylabel('Channel Bandwidth (kHz)')
title('Channel Bandwidth per User')
```

```
%% Objective 2: Data Rate
DataRate = CBW*2; % kbps
```

```
subplot(3,1,3)
bar(1:Users,DataRate*ones(1,Users))
grid on
xlabel('Users')
ylabel('Data Rate (kbps)')
title('Data Rate per User')
```

```
%% Objective 3: SNR Calculation
SignalPower = [mean(s1.^2) mean(s2.^2) mean(s3.^2)];
NoisePower = 0.01;
```

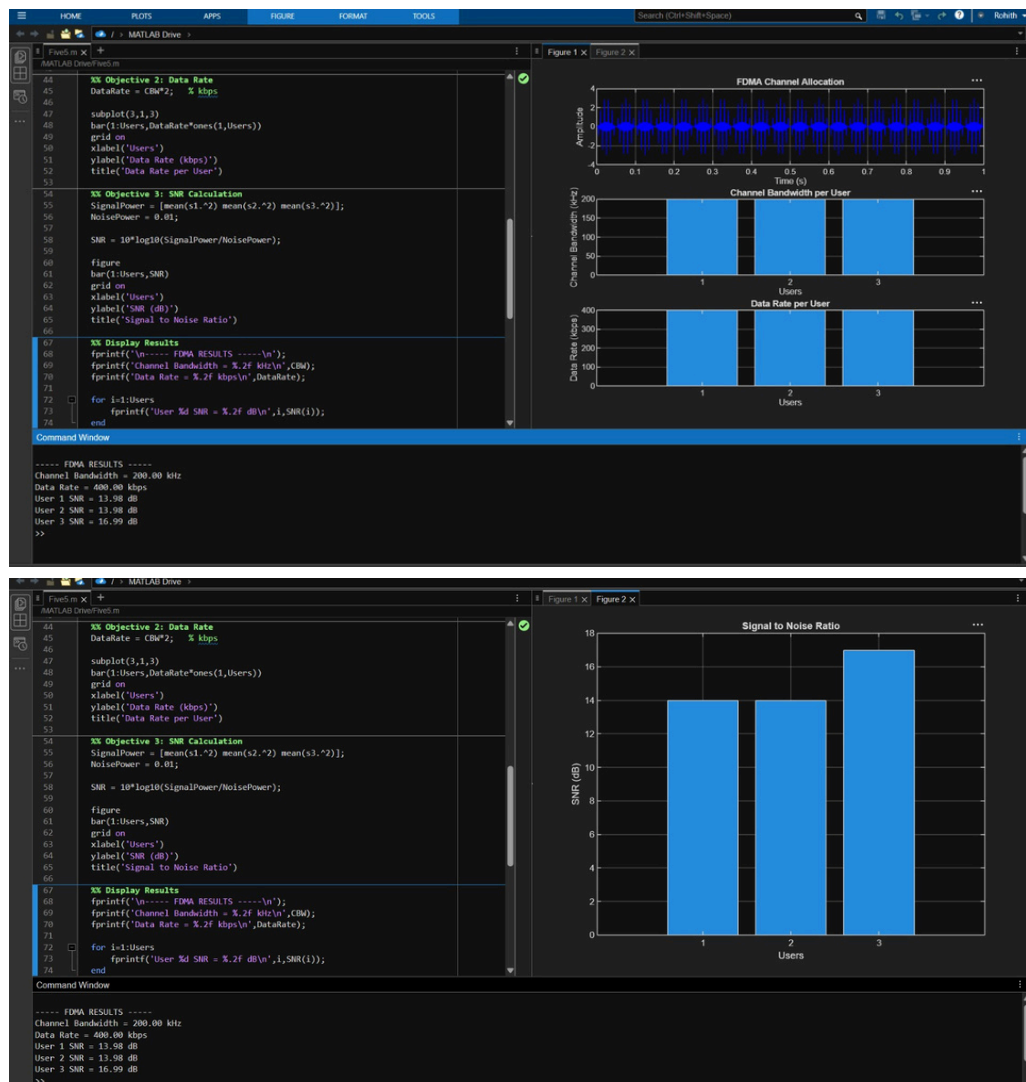
```
SNR = 10*log10(SignalPower/NoisePower);
```

```
figure
bar(1:Users,SNR)
grid on
xlabel('Users')
ylabel('SNR (dB)')
title('Signal to Noise Ratio')
```

```
%% Display Results
fprintf('\n----- FDMA RESULTS ----- \n');
fprintf('Channel Bandwidth = %.2f kHz\n',CBW);
fprintf('Data Rate = %.2f kbps\n',DataRate);
```

```
for i=1:Users
    fprintf('User %d SNR = %.2f dB\n',i,SNR(i));
end
```

6. Output Graphs



7. Result

The Frequency Division Multiple Access (FDMA) system was successfully simulated using MATLAB. Individual frequency channels were allocated to three users, and the channel bandwidth, data rate, and Signal-to-Noise Ratio (SNR) were calculated successfully. The output graphs confirmed proper channel allocation and demonstrated the performance of the FDMA system without interference between users.

8. Conclusion

The hybrid beamforming simulation demonstrates that hybrid beamforming provides a good balance between high throughput and reduced hardware complexity. The performance depends significantly on the number of RF chains, beam direction, power allocation, and SNR. Proper beam alignment and optimized power allocation improve throughput. Thus, hybrid beamforming is an effective technique for 5G and mmWave wireless communication systems, achieving performance close to fully digital beamforming with fewer RF chains.

This report follows the same structure and level of detail as your uploaded **AT2** report.